



Original research

The effects of visual feedback disruption on postural control with chronic ankle instability

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ABSTRACT

Objectives: The purpose of this study was to identify the effects of reduced visual feedback via stroboscopic glasses on dynamic postural control among chronic ankle instability (CAI), lateral ankle sprain (LAS) coper and uninjured control participants.

Design: Controlled trial in a laboratory setting.

Methods: Twenty CAI patients, 20 copers, and 20 controls participated in this study. Each participant performed a single-leg hop stabilization test with eyes open (EO) and stroboscopic vision (SV). Two-way ANOVAs (group $\times$ condition) were used to examine the differences between group (CAI, coper, and control) and condition (EO and SV).

Results: There was a significant group by condition interaction for DPSI scores. CAI patients displayed increased DPSI scores with SV compared to EO ($p < 0.01$), and CAI patients had increased DPSI scores only with SV when compared with controls. All participants displayed decreased dynamic postural control under the SV condition compared to the EO condition ($p < 0.01$) regardless of ankle group.

Conclusions: CAI patients rely more on visual feedback during dynamic postural control than copers and controls. However, they may not be as able to compensate for the disrupted visual feedback during a dynamic task. Regardless of ankle injury history, stroboscopic glasses could be cost-effective visual disruption devices during a dynamic balance task.

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Practical implications

- Stroboscopic glasses could be effective visual perturbation devices during a dynamic balance task regardless of ankle condition.
- CAI patients tend to rely more on visual information than copers and controls during dynamic postural control tasks
- Rehabilitation with stroboscopic glasses would help CAI patients to upregulate reliance on the somatosensory system and decrease the reliance on visual feedback during functional movements.

1. Introduction

Lateral ankle sprains (LASs) are one of the most frequent musculoskeletal injuries among physically active populations.¹ > 70% of LAS patients have experienced ongoing symptoms (e.g., swelling, pain, weakness, or decreased range of motion) for 7 years after their first-

time LAS.² Up to 75% of initial LAS patients suffer from these persistent symptoms³ along with decreased self-reported function and repetitive giving-way episodes, which can lead to a chronically unstable condition defined as chronic ankle instability (CAI). CAI may result from interrelationships among pathomechanical, sensory-perceptual, and motor-behavioral impairments.⁴ Despite a history of initial LAS, some individuals can return to pre-injury levels of activity without residual symptoms, self-reported dysfunction, or a feeling of giving way, and they are referred to as 'copers'.^{5,6} Understanding differences among CAI, copers, and control participants would elucidate not just factors contributing to CAI, but also protective strategies in copers to prevent further injuries.

It is well established that individuals with CAI displayed decreased dynamic postural control in a variety of tasks, such as the dynamic postural stability index (DPSI),⁷ and the star excursion balance test.⁸ However, it is still unclear how exactly sensorimotor dysfunction contributes to the deficits in postural control in CAI patients. Maintaining postural control is a combination of the appropriate efferent neuromuscular control system (e.g., increased muscular activity of the lower extremity for controlling upright stance)⁹ and the integration of the sensory information from visual, vestibular and somatosensory systems.¹⁰ Particularly,

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the interplay between somatosensory and visual feedback plays a significant role in providing appropriate afferent information to the central nervous system (CNS) to manage neuromuscular control during movements.¹¹ When one or more of these sensory information sources is altered, the CNS can shift its reliance to more reliable information sources among these sensory systems to stabilize posture; this is called sensory reweighting.¹⁰

To compensate for alterations in afferent information from the injured joint, individuals with musculoskeletal pathologies decrease the utilization of the somatosensory system and reweight more reliable sensory information.^{11,12} Specifically, individuals with multiple sclerosis have increased reliance on visual information during postural control.¹³ Moreover, individuals with anterior cruciate ligament reconstruction displayed increased activation in the posterior inferior temporal gyrus of the brain which is responsible for visual processing.¹¹ CAI patients also have shown greater visual reliance¹² and decreased ability to dynamically reweight the remaining sensory information when the somatosensory signals are altered via textured shoe insoles or through ankle joint anesthesia.¹⁴ However, there is still limited evidence of sensory reweighting in CAI patients.

In order to demonstrate the ability to successfully reweight visual feedback, clinicians or researchers have simply modified surfaces with foam pads¹⁵ or textured insoles¹⁶ to alter somatosensory information input. However, due to the lack of clinically implemented methods to disrupt visual feedback, assessing the reliance on vision has been limited to two extreme conditions: with and without vision. Recently, the development of novel technology, such as stroboscopic glasses, has made possible a visual disruption condition, which can be applied even during dynamic maneuvers.¹¹ Previous research¹⁵ demonstrated the effects of visual perturbation on static postural control using stroboscopic glasses. It suggested that a condition with stroboscopic glasses could provide an intermediary between the two extreme visual conditions during static postural control.¹⁵ However, the results were limited because static postural control cannot reflect the sensory reweighting of visual information during functional movements.¹⁵ In addition, sensory reweighting in healthy participants cannot be applied to CAI patients.

Since multiple studies have demonstrated the positive effects of stroboscopic glasses on conditioning, including skill training and sporting performance in various populations (e.g., normal students, ice hockey, baseball, etc.),^{17,18} the implications of these devices need to be extended beyond conditioning and into clinical practice, such as injury rehabilitation. If the condition with stroboscopic glasses could induce the sensory reweighting in CAI patients during dynamic balance, clinicians and researchers would have an easy and cost effective device to observe visual reliance during dynamic tasks. Therefore, the purpose of this study was to identify the effects of reduced visual feedback via stroboscopic glasses on dynamic postural control among CAI, copers and uninjured control participants. We hypothesized that CAI patients would demonstrate decreased dynamic postural control in the SV condition compared to copers and controls, based on their increased reliance on visual input.

2. Methods

A total of 60 physically active individuals (20 per group) with an age range of 18 to 35 years volunteered (Table 1). A priori sample size was calculated based on data from previous study investigating postural control among three groups.¹⁹ Approximated for two-way repeated measures ANOVA, a required sample size of 18 participants per group was estimated in order to reveal a group by condition interaction ($\alpha = 0.05$; $\beta = 0.2$). All participants in the three groups were matched for age (within 2 years), weight (within 5%), height (within 5%), sex and involved limbs. Inclusion criteria for CAI and copers groups were based on the recommendation by the International Ankle Consortium.²⁰ Participants were identified using the Foot and Ankle Ability Measure (FAAM)²¹ and the Ankle Instability Instrument

Table 1
Participant demographics.

	CAI	Coper	Control
N	10 M, 10 F	10 M, 10 F	10 M, 10 F
Age (years)	23.6 (3.9)	22.2 (1.4)	22.6 (2.7)
Height (cm)	174.5 (11.8)	176.4 (10.2)	174.4 (7.2)
Mass (kg)	76.3 (17.5)	69.3 (10.9)	75.7 (15.8)
FAAM-ADL (%)	83.04 (5.7)	100	100
FAAM-Sports (%)	64.22 (14.2)	100	100
All (No. Yes)	6.4 (1.1)	0	0

Data are mean (SD).

ADL, active daily living; All, ankle instability instrument; CAI, chronic ankle instability; F, female; FAAM, foot and ankle ability; M, male; N, number of participants.

(All)²² questionnaires. Specific inclusion criteria for each group is presented in Table 2.⁶ For the CAI patients who reported bilateral LASs, the injured ankle with the lowest FAAM scores was designated as the involved limb for the testing. Participants exclusion criteria were; (1) a history of surgery, (2) a history of fracture in the lower extremity, (3) a history of neurologic disorders, and (4) acute injury to musculoskeletal structures of lower extremity joints in the past 3 months. Prior to participation, all participants provided written informed consent prior to participation as approved by the Brigham Young University Institutional Review Board (Human Research Ethics Committee approval number: X18418). Supplementary Fig. 1 details the study flow from participant recruitment to data analysis.

All participants dressed with shorts, shirts, and athletic shoes (T-Elite XI; Nike, Beaverton, OR). Anthropometric data including height, mass, leg length, and age were recorded for each participant. Dynamic postural control was measured using an in-ground force plate (200 Hz, AMTI Corp., Watertown, MA) during a single-leg hop stabilization task.²³ To begin the jump-landing protocol, each participant stood 70 cm from the center of the force plate and was instructed to jump with both limbs to a height at which she or he could touch 50% of their maximum jump height target with a single hand and land in the center of the force plate. The participants were instructed to land in the center of the force plate on only the involved leg, stabilize as quickly as possible, and balance while standing erect and looking straight forward with the hands on their hips for 10 s.²³ Up to three successful trials of practice under each condition were allowed for each participant to decrease learning effects before actual data collection. Each participant performed three trials of the jump-landing test with eyes open (EO) and stroboscopic vision (SV) with a rest period of at least 60 s between trials. The order of conditions was randomized between participants. For the SV condition, we used stroboscopic glasses (Senaptec, Beaverton, OR) at 3 Hz (0.1 s of clear and 0.233 s of opaque). A trial was discarded and repeated if the participant lost their balance or did not touch the target.

During all testing conditions, postural control data were collected at a sample rate of 200 Hz using an in-ground force plate.²³ Dynamic postural stability data were reduced and calculated using a custom-written Matlab code (Version R2019b, MathWorks Inc., Natick, MA). For dynamic postural control, DPSI was calculated using GRF data from the force plate. Since the DPSI is a composite of the ML, AP, and vertical GRFs, stability indices for three directions were also calculated by a previous study's methods.²³ Three directional indices are mean square deviations assessing fluctuations around a zero point, rather than SDs assessing fluctuations around a group mean. The MLSI and APSI assess the fluctuations from 0 along the frontal and sagittal axes of the force plate, respectively. The VSI assesses the fluctuations from the participant's body weight to standardize the vertical GRF along the vertical axis of the force plate.

All data analyses were performed using JMP Pro 14 software (SAS Institute Inc., Cary, NC). One-way analysis of variance (ANOVA) were used to determine if differences in participant demographics including age, height, mass, FAAM-ADL, FAAM-Sports, and All existed among groups.

Table 2
Specific inclusion criteria for each group.

CAI	Coper	Control
1. ≥ 2 acute LASs required immobilization and/or non-weight bearing for ≥ 3 d's or external supports for ≥ 7 d's or both.	1. ≥ 1 acute LASs required immobilization and/or non-weight bearing for ≥ 3 d's or external supports for ≥ 7 d's or both.	1. No history of previous LAS.
2. History of at least two giving way episodes within past six months.	2. Return to moderate levels of weight-bearing physical activity without repeated ankle injury within past 12 months.	2. FAAM-ADL = 100%
3. FAAM-ADL < 90%	3. FAAM-ADL = 100%	3. FAAM-Sports = 100%
4. FAAM-Sports < 80%	4. FAAM-Sports = 100%	4. No "yes" answers on the All.
5. ≥ 5 "yes" answers on the All.	5. No "yes" answers on the All.	5. Physical activity ≥ 3 d's/wk. for 90 min within the past 3 m
6. No LE surgery and/or fracture.	6. Had not attended previous rehabilitation for the involved limb.	
7. Physical activity ≥ 3 d's/wk. for 90 min within the past 3 m.	7. Physical activity ≥ 3 d's/wk. for 90 min within the past 3 m	

ADL, active daily living; CAI, chronic ankle instability; d's, days; FAAM, foot and ankle ability measure; m, month; min, minutes; All, ankle instability instrument; wk, week;

All data were inspected from normality using the Shapiro-Wilk test and normal Q-Q plots. Two-way repeated measures ANOVAs were used to examine differences between condition (EO and SV) and group (CAI, coper, and control). The Tukey HSD post-hoc tests were used when the omnibus F p -value was < 0.05 . In addition, Cohen d effect sizes and 95% CIs were calculated to estimate the magnitude of visual condition effects in each group.

3. Results

The means, standard deviations, and effect sizes for each dynamic postural control measure by group and condition are presented in Table 3. There was a significant group by visual condition interaction for the DPSI scores ($p = 0.02$). However, there were no significant interactions for the other dynamic postural control measures (e.g., APSI, MLSI, and VSI). Post hoc analysis indicated that only CAI patients displayed increased DPSI scores with SV compared to the EO condition ($p < 0.01$). In addition, while CAI participants had no differences in DPSI scores with EO relative to copers and controls, they had increased DPSI scores compared with controls with SV ($p < 0.01$). There were significant group effects for DPSI ($p < 0.01$) and MLSI scores ($p < 0.01$). CAI patients had worse DPSI scores than controls ($p = 0.02$) and worse MLSI scores than both controls ($p < 0.01$) and copers ($p < 0.01$). Visual condition main effects ($p < 0.01$) were driven by differences in DPSI and MLSI scores between EO and SV. These results indicate that all participants had increased DPSI and MLSI scores ($p < 0.01$) with SV compared to the EO condition regardless of ankle group. The specific collapsed values for main effects for group and condition are presented in Supplementary Table 1.

4. Discussion

A stroboscopic visual disruption to decrease the dependence on visual feedback during dynamic postural control provides a novel

approach to investigating the sensory reweighting of visual feedback in individuals with CAI. The primary finding of this study was that the stroboscopic vision condition increased the DPSI scores in CAI patients relative to controls ($p < 0.01$). Moreover, only CAI patients displayed decreased dynamic postural control with SV compared to the EO condition. In contrast to the CAI group, copers displayed no differences in dynamic postural control during SV compared with controls, suggesting that copers may have the coping mechanisms to maintain balance. As such, copers may be able to effectively reweight their remaining sensory information when the visual information is limited. The secondary finding of the current study was that stroboscopic glasses alter performance in all three groups regardless of ankle injury status. Thus, stroboscopic glasses can be an effective rehabilitation and/or training equipment to provide the visual disruption condition during functional movements.

To our knowledge, the current study is the first to comprehensively examine the effects of visual feedback disruption on dynamic postural control among CAI, coper, and control participants. Previous research¹⁵ identified the fact that altered visual feedback via stroboscopic glasses decreased static postural control among healthy participants. However, because only uninjured healthy participants were enrolled in the previous study,¹⁵ we cannot apply their sensory reweighting findings to other pathological conditions such as CAI patients. In addition, data were limited to static postural control,¹⁵ which may be less applicable to dynamic tasks. Therefore, the results of current study provide a novel contribution toward understanding how CAI, coper, and control participants reweight their remaining sensory information to maintain dynamic postural control when the visual information is limited.

Current data partially confirm the hypothesis that dynamic postural control with SV could observe the differences in visual reliance between CAI patients and controls. Specifically, relative to controls, only CAI patients showed deficits only during SV conditions, but no differences under the EO conditions. Once peripheral mechanoreceptors around the foot and ankle joints are damaged from LASs, absent or inaccurate

Table 3
Values for the dynamic postural control measure by group and visual condition.

		CAI	Effect size (95% CI)	Coper	Effect size (95% CI)	Control	Effect size (95% CI)	Main effect group		Main effect visual condition		Interaction	
		Mean (SD)		Mean (SD)		Mean (SD)		F	p-Value	F	p-Value	F	p-Value
DPSI	EO	0.47 (0.05)	— 1.15	0.47 (0.06)	— 0.66	0.46 (0.04)	— 0.16	4.24	$< 0.02^b$	20.89	$< 0.01^d$	3.98	0.02 ^a
	SV	0.52 (0.03) ^e	(— 1.79, — 0.52)	0.50 (0.03)	(— 1.28, — 0.03)	0.47 (0.05) ^f	(— 0.78, 0.46)						
APSI	EO	0.31 (0.04)	0.00	0.30 (0.03)	— 0.64	0.30 (0.02)	— 0.02	1.07	0.35	1.81	0.18	1.65	0.20
	SV	0.31 (0.02)	(— 0.62, 0.62)	0.32 (0.03)	(— 1.26, — 0.01)	0.30 (0.03)	(— 0.64, 0.60)						
MLSI	EO	0.15 (0.05)	— 0.41	0.11 (0.04)	— 0.29	0.11 (0.04)	— 0.27	9.62	$< 0.02^{b,c}$	7.16	$< 0.01^d$	0.13	0.88
	SV	0.17 (0.04)	(— 1.03, 0.21)	0.13 (0.03)	(— 0.91, 0.303)	0.12 (0.03)	(— 0.89, 0.35)						
VSI	EO	0.34 (0.05)	0.11	0.34 (0.05)	0.02	0.33 (0.04)	0.04	0.75	0.48	0.37	0.55	0.11	0.90
	SV	0.34 (0.04)	(— 0.73, 0.51)	0.34 (0.05)	(— 0.63, 0.60)	0.33 (0.04)	(— 0.66, 0.58)						

^a Indicates a statistically significant group by visual condition interaction ($p = 0.02$).

^b Indicates a statistically significant difference between CAI and Control regardless of the visual condition ($p < 0.02$).

^c Indicates a statistically significant difference between CAI and Coper regardless of the visual condition ($p < 0.02$).

^d Indicates a statistically significant difference between EO and SV regardless of the group ($p < 0.01$).

^e Indicates a statistically significant difference between EO and SV within a group ($p < 0.01$).

^f Indicates a statistically significant difference between CAI and Control in SV condition ($p < 0.01$).

afferent information from the somatosensory sensory system may result in neuromuscular control deficits (e.g., postural control).^{24,25} To sustain and/or restore neuromuscular control in individuals with CAI, the CNS may compensate with increased reliance on visual feedback,¹² which is defined as sensory reweighting.¹⁰ The differences between EO and SV conditions which occurred only in the CAI group in this study may be attributed to the aforementioned altered sensory reweighting mechanism. In other words, current data support the idea that CAI patients cannot successfully reweight other sensory information¹⁶ and/or their impaired somatosensory system may limit efficient postural control when the visual information is altered.^{26,27} Therefore, we postulate that rehabilitation under the SV condition could be an effective method in clinical settings to potentially restore the ability of the CAI patients to reweight their remaining sensory information during dynamic postural control.

Current data support the idea that CAI patients have impaired dynamic balance compared with uninjured controls.²⁸ Particularly, only CAI patients displayed greater DPSI scores than controls (but not copers), which were consistent with previous research on dynamic balance between individuals with and without CAI.^{7,23} Increased DPSI scores in CAI patients may be due to the inadequate somatosensory input from damaged ankle joint after LASS,²⁹ whereas copers may restore their normal neuromuscular function to prevent further injuries. Interestingly, only MLSI scores of the three directional indices were worse in CAI patients relative to controls and copers regardless of the visual conditions. Previous studies^{5,27} recommended that ML GRF data is the most accurate measurement for the force plate in identifying balance deficits related to CAI; therefore, MLSI scores during a single-leg hop stabilization test are more accurate than APSI and VSI scores at distinguishing among the three groups.⁵ However, since the current method that requires the use of high-cost force plate equipment is not readily available in clinical settings, researchers need to examine what other low-cost devices are available to detect similar outcomes.

Our data were consistent with a previous study¹⁵ that the stroboscopic glasses provided a mechanism of visual disruption during postural control. Specifically, the ability to maintain dynamic balance decreased with SV compared with EO ($p < 0.01$). Although the visual reliance during postural control can be postulated by differences between EO and eyes closed conditions, comparing two different conditions is limited to only investigating the sensory substitution in which vestibular and somatosensory systems contribute to postural control without visual information.¹¹ However, utilizing disruption of the visual feedback condition may more closely simulate neurocognitive demands of daily living and sporting activities rather than using removed visual feedback conditions. In addition, it may demonstrate how individuals effectively reweight the remaining sensory information relative to injurious situations via visual disruption. Prospective studies should investigate how different sensory reweighting affects recurrent injuries among CAI patients.

It was also interesting to note that copers displayed similar postural control ability compared controls regardless of visual condition. That being said, copers may have different strategies to deal with limited visual information during dynamic balance tasks after their initial LAS, which may be a positive neuromuscular adaptation. Therefore, our results support the idea that copers do not heavily rely on visual feedback,³⁰ and may rather have greater movement strategy variability to compensate effectively for damage following an initial LAS.²³ Future research is needed to investigate the effect of visual disruption on functional movement strategies such as joint kinematics and EMG activation among individuals with and without CAI. However, as copers failed to show their superior ability in dynamic balance compare to CAI patients under the SV condition, positive neuromuscular adaption in copers may be less effective than strategies used by controls.

Although it remains unclear how intermittent visual feedback exactly affects the mechanism of sensory reweighting, stroboscopic

glasses allowed all participants to dynamically reweight the remaining sensory information to maintain postural control. However, as all participants showed worse DPSI and MLSI scores under the SV condition compared to the EO condition regardless of ankle group, other residual sensory information sources (e.g., vestibular, somatosensory) may not provide sufficient afferent information to fully compensate for the limited visual feedback during dynamic postural control. Therefore, stroboscopic glasses could be cost-effective visual disruption devices that can provide neurocognitive demanding condition during functional movements for a physically active population. Furthermore, these devices may be beneficial in upregulating reliance on somatosensory information for motor programming. Future research is needed to investigate the effects of rehabilitation and/or training with SV on sensory reweighting strategies during functional movements or postural control in individuals with altered somatosensory input due to musculoskeletal injuries.

There are several limitations in the current study. First, the population of this study was physically active and young adults. The current findings can be generalized to only the particular populations. Second, due to the limitation of a cross-sectional design, it is unclear whether CAI and copers participants had altered visual reliance prior to their LASS or whether alterations are caused by ankle injury. Third, all postural control abilities were interpreted by the GRF based data. Future research is needed to identify altered joint kinematics and musculature activation to compensate for limited visual information during the tasks.

5. Conclusion

In conclusion, CAI patients rely more on visual feedback during dynamic balance than copers and controls. However, copers displayed no differences regardless of visual condition compared to controls, which may be a positive neuromuscular adaptation following an initial LAS. Regardless of ankle injury history, stroboscopic glasses could be effective visual disruption devices during a dynamic balance task. However, across all groups, other remaining sensory information (e.g., vestibular, somatosensory information) may not provide sufficient afferent information to overcome the reduced visual feedback during dynamic postural control.

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Declaration of interest statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Confirmation of ethical compliance

Institutional Review Board (IRB) approval was granted for all procedures.

Consent to participate. Written informed consent was received from all participants following IRB guidelines and approvals.

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